

to other ships. Each individual ship receives data and fuses it with its own radar data. This is carried out by employing high-capacity parallel-processing computers and advanced fusing algorithms. The result is that, all individual ships see a common air defence scenario called air picture.

Earlier a destroyer or a cruiser could view the threat scenario on the air around it. But through CEC, the same vessel can now see the threat scenario on air around the entire CSG.

The backbone of CEC is a robust communications system, which is jam-resistant and utilises the advantages of GPS. Ships communicate in pairs in 500MHz to 1000MHz band during short transmit/receive periods to a line-of-sight (radar) distance. Data thus sent across the net is in near real-time, and communication is virtually jam-proof because of superior jam-resistant features.

The single greatest advantage is that a ship may not be seeing an incoming cruise missile through its radar, but can launch and guide a SAM to intercept the incoming cruise missile based on fused track data sent by other ships. Similarly, a ship may launch a SAM with another ship providing midcourse guidance with yet another ship illuminating the target.

For example, a single ship's radar may be able to detect a cruise missile at 40km from the ship. If the cruise missile is travelling at a speed of 1mach (330m/s), then it will reach the ship in 120 seconds. So the ship has to intercept the incoming missile within two minutes to prevent sinking.

Assume that there are two CEC equipped ships, A and B, operating 40km apart on a straight line. Ship A detects a cruise missile at 40km from it along its axis and the cruise missile is flying towards ship B. Through CEC, as ship A detects the cruise missile, ship B also detects it. Now, the cruise missile has been detected at 80km from ship B against its 40km-detection capability. Due to this, ship B now sufficiently has four minutes

to intercept the missile. This feature greatly relieves the pressure on ship B, when it is operating near the shores.

With CEC, each individual ship is made to scan a specific sector and data from all ships is fused into a single air picture. Interestingly, two CSGs operating in close proximity can share the data. This enables both to have an overall air picture common for both strike groups.

Such capabilities have been achieved for the first time in history and even been deployed operationally.

Phase IV: Terminal air defence

This phase is almost a cruise missile defence phase executed by the carrier, but can also tackle enemy aircraft, if at all enemy aircraft survive the previous air defence phases, which is a near impossible feat. But with cruise missiles, it is a different story.

Cruise missiles fly so low as if these are skimming over water surface. This flight profile helps these to hide among clutter, that is, echoes from the sea surface. With this flight profile, these may still escape interceptions carried out by Aegis vessels and reach the carrier, generally considered remotely possible. An enemy aircraft or cruise missile reaching this phase means that Aegis vessels have been neutralised.

For the carrier, there are two defensive lines against cruise missiles. One is through Sea Sparrow SAM system and the other is through ship self defence system (SSDS). AN/APS-49 and AN/APS-48 radars of the carrier provide coordinates of the targets in the initial phase of engagement.

Evolved Sea Sparrow SAM system. Evolved Sea Sparrow is the name given to SAM. The entire system consists of a target acquisition radar called Mark-23 target acquisition system (TAS), an illuminator radar and missile launchers, all tied up to a fire control system (FCS) called Mark-91.

FCS is the point of man-machine interface. It is this FCS which has the FIRE switch to launch the missile.





PG08



PG09H

PG08 (Max. 10A)

- Power (W)
- Power Factor (PF)
- Current (mA)
- Voltage (V)
- Frequency (Hz)
- Total Usage Time (min)
- Total Used Energy (KWh)
- Carbon Emission (CO₂)

PG09H (Max. 15A)

- TRMS Voltage (V)
- TRMS Current (mA)
- Power (W)
- Energy Usage Time (EUT)
- Energy Consumption (KWh)

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